Setting Up a Docker Container

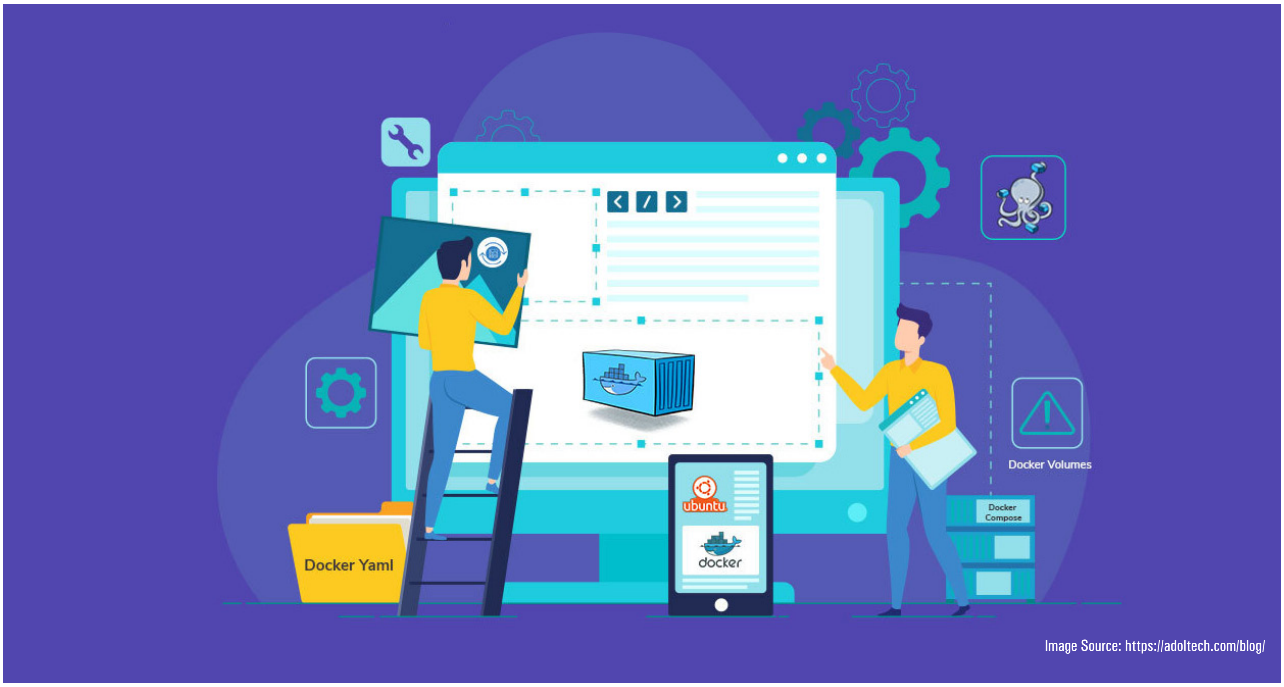

Image Source: https://adoltech.com/blog/

Docker provides a lightweight virtualization solution with zero overhead. This article demonstrates how to install Docker on Ubuntu. It also outlines the deployment of a Docker container application, NGINX, on Ubuntu.

Containerization is currently popular in the software development community as an alternative to virtualization. It enables us to encapsulate (containerize) software with all its dependencies and the runtime environment so that it can run uniformly across various infrastructures and platforms. In general, running software in containerized environments consumes lesser computing resources than running software in multiple virtual machines, because each virtual machine requires its own copy of the operating system. This article compares a virtual machine to a Docker container and demonstrates how to install and configure a Docker containerization environment. Finally, deployment of NGINX as a sample Docker container application is also demonstrated.

Virtual machine vs Docker container

Docker is a software development platform enabling virtualization on a single host with different operating systems. It enables rapid software delivery by separating infrastructure from applications. Docker virtualization is performed at the system level, using Docker containers, unlike hypervisors, which produce VMs. As depicted in Figure 1, Docker containers operate the operating system of the host. This also improves an application's efficiency and security. Also, as containers consume fewer resources than virtual machines, we can deploy more containers on the same hardware.

A comparison of VM and Docker containers is given in Table 1.

Table 1: Comparison of virtual machine and Docker container

Virtual machine (VM)	Docker
Hardware-level isolation	OS-level isolation
Slow booting (minutes)	Fast booting (sec)
Heavyweight (Gigabyte)	Much smaller in size than a VM (KB/ MB)
VMs can simply migrate to new hosts	Containers are destroyed and rebuild
Huge computing resources required	Less computing resources required